Engineering Mechanics 1 · Unit_01 · UNIT 1

DRAFT — UE26ME142A · Unit_01 · UNIT 1 · UE26ME142A Unit 1 Class 3 AVSummary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 1 Class 3 AVSummary

1 UE26ME142A Unit 1 Class 3 AVSummary

Study focus: UE26ME142A_Unit 1_Class 3_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000086 · EV0000087

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Svink Tish, Civil Engineering, Department of Science and Humanities, PS University.
- This is class number six.
- In the last class, we had discussed about the a difference between components and projection.
- How do we determine the components?
- How do we determine the projection?
- What is rectangular?
- And several definitions that we were supposed to know to solve problems of engineer mechanics.
- In this class we will look at what are the problems that we can solve.
- We will try to solve one unsolved problem also.
- The statement of the problem is the forces F1, F2, and F3, all of which act on point A of the bracket.
- Are specified in three different ways.
- Determine the X and Y scalar components of each of these three forces.
- And what is asked is to determine the X and Y scalar components, only the components of the forces along X and Y.
- This is point A.
- He says three different ways.
- The direction of the forces is specified in three different ways.
- What are these three different ways?

- One is for.
- They have mentioned the angle made by the force with respect to the x axis.
- X axis is horizontal, y axis is vertical.
- This we have to make out from the diagram.
- Or this says a line which is parallel to X axis.
- The vertical distance is three, the horizontal distance is four.
- Or we can also determine what is the angle made by F2 with respect to x axis or y axis.
- Say with respect to x axis, it will be equal to tan inverse of 3 by 4.
- The direction of the force F3 is not mentioned.
- Instead, they have mentioned the starting point of F3, the ending point of F3 as B, A and B, starting point is A, ending point is B.
- from the origin as point two along x axis.
- So first force F1 is indicated here.
- This is F1 which makes an angle of thirty five degrees with the x axis.
- It should be equal to F cos 35 degrees straight away and F1y will be equal to F1 sine 35 degrees.

Example or application from the lecture

- I've taken a few example problems to understand how do we go about solving these problems and let's see depending upon the time.
- We should read the diagram that is given.
- From the statement of the problem, what is given is there are three forces which is acting at point A, that we can make out from the diagram.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

Core explanation · 05:00–09:59

To determine F2x, if I take this.

Key details

- It is gonna be equal to F two cause of this angle, cause of this angle is 4 divided by 5 into 500 will give me F2x.
- F3 is acting here.
- Now if I take that angle made by the force F3 with respect to y axis as alpha, I can easily calculate what is the angle alpha as tan inverse of 0.2 by point four because this forms a right angle triangle.
- Adjustant opposite is 0.2, adjustant is 0.4, so I can get alpha.
- These are the scalar components of the forces F1, F2, and F3.
- which act on the fixed structure at B into a single equivalent force R.
- Resultant of the two forces that is force P and T, what is the resultant of these two forces?
- We can solve this in two different approaches, or whichever approach you want, you can take several approaches.

- Parallel of forces, the other one is using the law of triangle of forces, and one more is by resolving P under T.
- Along X and Y, and then combining out finding out the dot product of this, you will get the result.
- You can see that P is represented here, which is 800 newtons.
- forms or this two forces are represented now on the adjacent sides of a parallelogram.
- Both of them are moving away from point B.
- And how do I find this magnitude of the resultant?
- That would be equal to P square.
- Plus T square plus 2 times P into T into cos of the angle made by these two.
- How do I find the angle?
- or rather 180 minus alpha.
- I can find out what is the result and r from that.
- The two forces are taken on the adjacent sides of a triangle, taken in order, you can see that this is in order.
- The closing side of the triangle taken in the opposite order gives us the resultant.

Example or application from the lecture

- Now let us move on and take one more example.
- This example the statement of the problem states that combine the two forces P and T.
- Let's look at all these three different ways of solving the problem, one at each time.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 10:00–14:59

The other way of solving this is by considering or finding out the component of the two forces along X and Y, and then finding out the resultant R.

Key details

- How do I get Rx square plus Ry square?
- I get the magnitude of R.
- R X coming this, R Y coming this.
- When I merge these two, I get the resultant force acting like this.
- This is how we can determine the resultant of the of these two forces.
- Vector approach to solve.
- In the two previous problems that I have explained, what we have looked at is how do we solve it by going ahead and solving it in scalar approach.
- The 500 newton force F is applied to the vertical pole as shown.
- This is a vertical pole.
- A 500 newton force that is this one is applied to this vertical pole as shown.
- This is acting at point A, which makes an angle of 30 degrees with the.
- Y-axis or with the pole, the forces are acting at an angle of thirty degrees.
- The magnitude of the force is 500 newtons.

- Write F in terms of the unit vectors I and J and identify both its vector and the scalar components means what?
- We have to find out the component of the force f along x and along y.
- How do I make out that?
- It is i and j.
- Then we have to determine both scalar and vector approach.
- Scalar means only Fx and Fy, we have to determine that.
- The second part of the problem is determine the scalar component of the force vector along the x-dash and the y-dash axis.
- Which is the x-dash axis?
- The axis along the force itself is x dash and perpendicular to that is y dash.
- You have to find out the component of the force along this x dash and y dash.
- That is second one.
- The third one is determine the scalar components of force.
- Yes, along the x and the x-dash axis, that means what?
- I have to find out the scalar component of this force along x and along y-dash axis, along this axis.
- That is what we have to find out.
- So first thing we have taken the first case where we have to determine the.
- A component of the force along X and Y.
- The component of the force is also represented along X and Y.
- Whereas F sine sixty degrees with a negative sign, I'll get it as F y.
- Why it is negative?
- Because the sense of the force F y is acting downwards, so I get it as negative.
- Now to express this as a vector, it is nothing but fx i minus f y j because f y is acting downwards, so it is negative.
- The scalar component is just fx which is equal to f cos 60 degrees.
- The scalar component of y that is f y is equal to f cos 60 degrees, that's it.
- The next one is to determine the component of the force along x dash and y dash axis.
- The component of the force along Y will be F, cos ninety degrees, and we know that cos ninety degrees is.

Example or application from the lecture

- Let's go at and try to solve another problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

Core explanation · 15:00–19:59

Next coming to determination of the scalar component of the force f along x and y dash axis.

Key details

- Where you rotate it and merge it up to ninety less we can rotate it.
- Parallel of forces.

- They have asked you to determine the components.
- Component is when we have to determine by using law of triangle of forces.
- Component will produce the same effect as that of the force.
- We can determine the component of F along F x and FY.
- For that, we have to determine the angles.
- This is parallel to this, so this forms the same angle, 60 degrees and 60 degrees.
- Ah whereas these two will be the same, so they are alternate angles, so 90 and 90.
- We know that this forms a triangle.
- by sine 90 is equal to f pi sine 30.
- By sine 60 will be equal to f by sine 30, so I can get f y dash.
- The statement of the problem is forces F1 and F2 act on the bracket as shown.
- Determine the projection of FP of their resultant R onto the B axis.
- Clearly understand the problem.
- What is asked is there are two.
- This is axis A, this is axis B.
- They have asked you to determine the projection of these the resultant of these two forces.
- First we have to find out what is the resultant of these two, then projection of the resultant onto the B axis.
- How do I construct it?
- We have this force and this force.
- We know this angles.
- The total angle is 50 degrees.
- That is 20 plus 30 is 50.
- Is along A and this force F2 is along B.
- This is parallel to this.
- this edge, so these two lines when I draw, it was going to cross each other at this point.
- The projection of R onto the B axis.
- Touches B axis such that this angle is at 90 degrees.
- That would be equal to F P plus this value.
- This value will be equal to F1, which is same as this.
- Cause of this will be this value.

Example or application from the lecture

- Forces F1 of magnitude 100, F2 of magnitude 80 newtons, the sense of the two is given, the direction of the forces are also given as 30 degrees with respect to X, 20 degrees also with respect to X.
- Let's look at this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

Core explanation · 20:00–24:59

Which is equal to 144.3 newtons.

Key details

- That would be the projection of the resultant on the B axis.
- I have explained it here.
- F2 square plus F1 square plus 2 times F1 F2, cause of the angle between the two.
- This angle which is 50 degrees, I'm sorry I have made a mistake here.
- I have taken it as 130, it should be 50 degrees.
- I hope this is very clear.
- We will come back to this because I'm shooting the time.
- We will look at this problem in the next class.
- my name is Doctor Svang Tesh.
- We'll meet in the next class, class number seven.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 1_Class 3_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/003_UE26ME142A_Unit_1_Class_3_AVSummary__eee232e254abeb39.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 3_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/003_UE26ME142A_Unit_1_Class_3_AVSummary__eee232e254abeb39.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 1_Class 3_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/003_UE26ME142A_Unit_1_Class_3_AVSummary__eee232e254abeb39.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 1_Class 3_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_01/AV_Summaries/003_UE26ME142A_Unit_1_Class_3_AVSummary__eee232e254abeb39.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.0_Lecture_PPT__a53eb8b08a5ac7ff.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 001_UE26ME142A_Unit_1_Class_1.1_Lecture_PPT__a6f76a088910b54a.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.0_Lecture_PPT__de74ea957f21118c.slides.md have no readable `.slides.md` text; their original page images are included.
4. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.1_Lecture_PPT__bb478549c01cfb60.slides.md have no readable `.slides.md` text; their original page images are included.
5. 2 rendered pages in 002_UE26ME142A_Unit_1_Class_2.2_Lecture_PPT__3fc8761d12196c6f.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 003_UE26ME142A_Unit_1_Class_3_Lecture_PPT__40166a1710a03d41.slides.md have no readable `.slides.md` text; their original page images are included.
7. 3 rendered pages in 004_UE26ME142A_Unit_1_Class_4_Lecture_PPT__984b7c368532af06.slides.md have no readable `.slides.md` text; their original page images are included.
8. 3 rendered pages in 005_UE26ME142A_Unit_1_Class_5_Lecture_PPT__f49d361d79853022.slides.md have no readable `.slides.md` text; their original page images are included.

9. 2 rendered pages in 006_UE26ME142A_Unit_1_Class_6_Lecture_PPT__f79618b4a0f260c2.slides.md have no readable `.slides.md` text; their original page images are included.
10. 3 rendered pages in 008_UE26ME142A_Unit_1_Class_8_Lecture_PPT__2659a840cc2bfd13.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 009_UE26ME142A_Unit_1_Class_9_Lecture_PPT__832f697eaf1d383c.slides.md have no readable `.slides.md` text; their original page images are included.
12. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.0_Lecture_PPT__aa8f331eb00d810f.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 010_UE26ME142A_Unit_1_Class_10.1_Lecture_PPT__fe73ddc3ca596069.slides.md have no readable `.slides.md` text; their original page images are included.
14. 2 rendered pages in 011_UE26ME142A_Unit_1_Class_11_Lecture_PPT__3624142116c7273d.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 012_UE26ME142A_Unit_1_Class_12.0_Lecture_PPT__2e7b61b526d51f60.slides.md have no readable `.slides.md` text; their original page images are included.
16. 4 rendered pages in 012_UE26ME142A_Unit_1_Class_12.1_Lecture_PPT__1d60317fe4af0cbe.slides.md have no readable `.slides.md` text; their original page images are included.
17. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
18. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
19. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
20. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
21. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00022	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00023	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00025	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

C Revision and retrieval practice